

CHEMISTRY

Reaction:



Overview

Rocket "candy" is used as solid fuel for amateur rockets. It:

- Uses **sucrose** (sugar) as fuel
- **Potassium nitrate** as an oxidizer
- **Common metal oxides** (e.g. iron(III) oxide) to catalyze the reaction

- **Sucrose**: hydrocarbon, provides energy in combustion reaction ($2\text{C} + 2\text{H}_2 + 3\text{O}_2 \rightarrow 2\text{CO}_2 + 2\text{H}_2\text{O}$)
- **Potassium nitrate** releases oxygen in reaction
- **Metal oxide**: catalyst, ensure smoother combustion
 - irregularity creates pockets where gas can escape, preventing pressure build up

ROCKET CANDY

- **Dry compression**
 - Sugar & potassium nitrate individually grounded finely
 - Mixed into a ball mill/tumbler
 - Packed into a motor tube
- **Dry heating**
 - Potassium nitrate ground into fine powder
 - Mixed with sugar, then heated
 - Coats grains of KNO_3 with sugar

- **Dissolving & heating**
 - Sugar and potassium nitrate are heated in water
 - Sugar dissolves, coating the potassium nitrate crystals
 - Forms a paste of the melted sugar and potassium nitrate

PREPARATION

APPLICATION

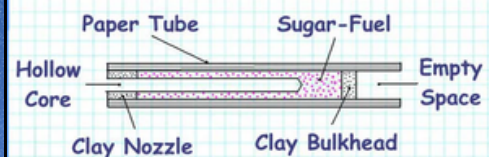
- First popularised in 'Rocket Manual for Amateurs' by Bertrand R. Brinley & described by Lt. Col. Charles M. Parkin in 'Electronics Illustrated'
 - Often used in amateur rockets
- Used as crude weapons of war
 - Attacks on Israel by Hamas, 2000-2003

- Sugar Shot to Space programme
 - Goal: loft a rocket powered by sugar propellant into space
 - Double Sugar Shot expected to reach 33km
 - First Mini Sugar Shot rocket reached 4km
- Safer for use for amateur rocketry as materials are non-hazardous

ROCKET CANDY



1/2" Sugar-Fueled Rocket Motor



PHOTOS